

Stellar Parameter Determination in Open Clusters: A Comparative Benchmark of 2D Bayesian Isochrone Fitting, Machine Learning Regressors, and Empirical Calibrations

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Abstract

Determining fundamental stellar parameters — such as mass ($M_{\odot}$), age (t), effective temperature (T_{eff}), surface gravity ($\log g$), bolometric luminosity ($\log_{10} L/L_{\odot}$), and stellar radius (R_{*}) — is essential for constraining star formation histories, IMF variations, and disk accretion physics across open clusters. In this work, we present an empirical benchmark comparing 2D Bayesian Isochrone Fitting (IsocFit) incorporating Salpeter/Kroupa IMF priors, semi-empirical Machine Learning Mass–Magnitude Relationships (RMM), and spectroscopic Pre-Main-Sequence (PMS) activity classifications across five open clusters spanning evolutionary stages from ~ 1 to ~ 150 Myr: Orion Nebula Cluster (ONC, ~ 1 Myr), Upper Scorpius (~ 5 – 11 Myr), h Persei / NGC 869 (~ 13 Myr), Pleiades (~ 112 Myr), and NGC 2516 (~ 150 Myr). Methodological validation against eclipsing binary control stars demonstrates that k -NN regressors trained on typical stellar evolutionary tracks reproduce empirical masses with 3.8% median residual uncertainty ($R^2 = 0.96$, homoscedasticity $P = 0.12$). For young PMS clusters (< 10 Myr), 2D Bayesian isochrone likelihood estimation accurately captures age dispersion and individual $1\text{-}\sigma$ posterior uncertainties, whereas k -NN ML models using *Gaia* DR3 G/V magnitudes minimize extinction and bolometric correction sensitivity. Conversely, for clusters near or beyond Zero-Age Main Sequence (ZAMS) convergence (> 100 Myr), k -NN regressors resolve track-folding degeneracies, yielding superior agreement with reference catalogues ($R^2 \approx 0.96$ – 0.99). Finally, incorporating spectroscopic Equivalent Widths ($H\alpha$, Li I, TiO) provides an robust boundary distinguishing Classical T Tauri Stars ($H\alpha \geq 10 \text{ \AA}$) from Weak-lined T Tauri and field stars. All algorithms are implemented in the open-source software SPECTRA (v1.2.0).

Keywords: stars: pre-main-sequence; stars: fundamental parameters; open clusters and associations: general; stars: low-mass; Hertzsprung–Russell and colour-magnitude diagrams

1. Introduction

Determining fundamental stellar parameters — such as mass ($M_{\odot}$), age (t), effective temperature (T_{eff}), surface gravity ($\log g$), bolometric luminosity ($\log_{10} L/L_{\odot}$), and stellar radius (R_{*}) — is central to observational and theoretical astrophysics [1]. Interior stellar structure [2], angular momentum evolution [3], magnetic dynamo action, and circumstellar disk dissipation timescales are all fundamental mass- and age-dependent processes.

In stellar clusters, coevality and homogeneous metallicity provide key leverage for placing stars on theoretical Hertzsprung–Russell (HR) diagrams or Colour–Magnitude Diagrams (CMDs).

However, estimating stellar masses and ages for pre-main-sequence (PMS) stars ($t \lesssim 100$ Myr) remains challenging due to several observational and theoretical complexities [4]:

1. High observational uncertainties caused by differential interstellar extinction (A_V) and accretion continuum veiling;
2. Real or apparent age spreads within star-forming regions;
3. Systematic divergences between independent theoretical evolutionary tracks (e.g., Siess et al. 5; Baraffe et al. 2) arising from differing treatments of convection, atmospheric opacity, and deuterium burning boundaries;
4. Convergence and folding of evolutionary tracks near the Zero-Age Main Sequence (ZAMS) for intermediate-mass and older stars ($t \gtrsim 100$ Myr), creating geometric degeneracies during HR diagram grid minimization.

To address these limitations, empirical and semi-empirical Mass–Luminosity and Mass–Magnitude Relationships (MLR/MMR) derived from double-lined eclipsing binaries [6,7] provide model-independent benchmarks. Furthermore, machine learning (ML) regression algorithms (e.g., k -Nearest Neighbours, Random Forests, Support Vector Regressors) trained on synthetic or empirical evolutionary grids offer a flexible alternative to traditional grid interpolation, mitigating sensitivity to bolometric corrections and reddening laws [8]. In addition, the development of full Bayesian inference frameworks [1] provides rigorous propagation of correlated uncertainties, replacing naive nearest-neighbour grid matching with posterior probability density functions (PDFs) that can formally incorporate priors on the Initial Mass Function (IMF) and star formation history.

In this paper, we present a comprehensive scientific benchmark evaluating 2D Bayesian Isochrone Fitting (*IsocFit*), machine-learning Mass–Magnitude relationships (RMM), and multi-band photometric/spectroscopic primary parameter derivations. We analyze five target stellar clusters covering a wide range of evolutionary ages: the Orion Nebula Cluster (ONC, ~ 1 Myr), Upper Scorpius (~ 5 – 11 Myr), h Persei / NGC 869 (~ 13 Myr), Pleiades (~ 112 Myr), and NGC 2516 (~ 150 Myr). All calculations and diagnostic workflows are implemented in the open-source software SPECTRA (v1.2.0).

This paper is structured as follows. Section 2 details the computational framework, 2D Bayesian estimator, ML regression pipeline, primary parameter derivations, and spectroscopic T Tauri activity classification. Section 3 presents validation results on an empirical eclipsing binary control sample. Section 4 details the application and comparative results across the benchmark clusters. Section 5 synthesizes methodological trade-offs, and Section 6 presents our primary conclusions.

2. Methodology and Computational Framework

The analytical framework developed in SPECTRA integrates four primary computational modules designed to process multi-catalog photometry (*Gaia* DR3, 2MASS, Johnson–Cousins) and spectroscopic feature indices.

2.1. 2D Bayesian Isochrone Parameter Estimator (*IsocFit*)

Traditional isochrone fitting methods often rely on 1D grid minimization or Euclidean distance minimization in HR space. However, such methods struggle to provide robust parameter uncertainties. In *IsocFit*, we implement a full 2D Bayesian posterior probability estimator [1].

Given an observed star with effective temperature $T_{\text{eff,obs}} \pm \sigma_T$ and logarithmic bolometric luminosity $\log L_{\text{obs}} \pm \sigma_L$, the joint Gaussian likelihood $L(d|M, t)$ over a theoretical grid model (M_i, t_j) is given by:

$$L(d|M, t) = \frac{1}{2\pi\sigma_T\sigma_L} \times \exp \left[-\frac{1}{2} \left(\frac{(T_{\text{eff,obs}} - T_{\text{eff,grid}}(M, t))^2}{\sigma_T^2} + \frac{(\log L_{\text{obs}} - \log L_{\text{grid}}(M, t))^2}{\sigma_L^2} \right) \right] \quad (1)$$

To account for the physical mass distribution of stars in clusters, we adopt an explicit Initial Mass Function (IMF) prior $P(M)$ based on the Salpeter/Kroupa power-law formulation [9,10]:

$$P(M) \propto \begin{cases} M^{-1.35}, & \text{if } M \leq 0.5 M_{\odot} \\ M^{-2.35}, & \text{if } M > 0.5 M_{\odot} \end{cases} \quad (2)$$

Assuming a uniform prior over logarithmic age $P(t) \propto 1$, the posterior probability distribution $P(M, t|d)$ is obtained via Bayes' theorem:

$$P(M, t|d) = \frac{L(d|M, t) P(M) P(t)}{\iint L(d|M, t) P(M) P(t) dM dt} \quad (3)$$

The expected stellar mass $\hat{M}$ and age $\hat{t}$ are calculated as the posterior expectation values:

$$\hat{M} = \iint M P(M, t|d) dM dt, \quad \hat{t} = \iint t P(M, t|d) dM dt \quad (4)$$

The 1- σ posterior uncertainties (σ_M, σ_t) are computed directly from the second central moments of the 2D marginal distributions:

$$\sigma_M = \sqrt{\iint (M - \hat{M})^2 P(M, t|d) dM dt} \quad (5)$$

Theoretical evolutionary tracks supported in `IsocFit` include those from Siess et al. [5] and BHAC15 [2]. The choice of these tracks is historically and physically motivated: Siess et al. [5] grids remain a heavily referenced standard for PMS evolution spanning a wide mass range, providing a long-standing baseline for comparing legacy studies, while the BHAC15 tracks incorporate updated solar abundances, atmospheric boundary conditions, and improved convection mixing-length calibrations, making them particularly well-suited for very-low-mass stars and brown dwarfs. Combining these two complementary grids ensures comprehensive coverage of both classical and modern evolutionary constraints.

2.2. Machine Learning Mass–Magnitude Relationships (RMM)

As an alternative to HR grid interpolation, `SPECTRA` builds semi-empirical Mass–Magnitude Relationships (RMM) by training supervised machine learning regressors directly over theoretical grids downloaded via the `MADYS` repository manager [8].

Supported regression algorithms include k -Nearest Neighbours (k -NN), Random Forests (RF), Support Vector Regressors (SVR), Gradient Boosting (GB), Ridge, Lasso, and Bayesian Ridge. Regressor hyper-parameters are optimized using K -fold cross-validation ($K = 5$) and Grid Search.

To rank candidate models objectively, `SPECTRA` evaluates a multi-metric composite score S :

$$S = \hat{\mathcal{M}} \cdot \mathcal{P}^T \quad (6)$$

where $\hat{\mathcal{M}}$ is the min-max normalized performance matrix containing Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), $(1 - R^2)$, and Akaike Information Criterion (AIC), and $\mathcal{P} = [0.25, 0.25, 0.25, 0.25]$ is a uniform weighting vector.

2.3. Primary Stellar Parameters and Bolometric Luminosity Derivation

For catalogs lacking pre-computed effective temperatures or luminosities, the Primary Parameters module derives fundamental physical quantities directly from multi-band photometry (*Gaia* G, BP, RP , 2MASS J, H, K , Johnson V, B).

Intrinsic dereddened colors are derived using Cardelli interstellar extinction ratios A_λ / A_V [11]:

$$m_{\lambda,0} = m_\lambda - A_\lambda = m_\lambda - R_\lambda A_V \quad (7)$$

Multi-color effective temperature estimates (T_{eff}), spectral types (SpT), surface gravities ($\log g$), and bolometric corrections (BC_λ) are interpolated using empirical dwarf tables [12].

Bolometric luminosity $\log_{10}(L_*/L_\odot)$ and stellar radius $R_*/R_\odot$ are derived strictly following Section 3.2 of Bell et al. [3]:

1. **Distance Modulus (DM):** Calculated from stellar distance d_{pc} or parallax ϖ_{mas} :

$$\text{DM} = 5 \log_{10}(d_{\text{pc}}) - 5 = 15 - 5 \log_{10}(\varpi_{\text{mas}}) \quad (8)$$

2. **Absolute Bolometric Magnitude (M_{bol}):**

$$M_\lambda = m_{\lambda,0} - \text{DM}, \quad M_{\text{bol}} = M_\lambda + BC_\lambda(T_{\text{eff}}) \quad (9)$$

3. **Logarithmic Bolometric Luminosity ($\log_{10}(L_*/L_\odot)$):**

$$\log_{10}(L_*/L_\odot) = 0.4[M_{\text{bol},\odot} - M_{\text{bol}}] \quad (10)$$

where $M_{\text{bol},\odot} = 4.755$ mag is the solar bolometric absolute magnitude adopted by Bell et al. [3].

4. **Stellar Radius ($R_*/R_\odot$):** Via the Stefan–Boltzmann law:

$$\log_{10}(R_*/R_\odot) = 0.5 \log_{10}(L_*/L_\odot) - 2 \log_{10}\left(\frac{T_{\text{eff}}}{5770 \text{ K}}\right) \quad (11)$$

2.4. Spectroscopic EW Diagnostics and T Tauri Classification

To diagnose Pre-Main-Sequence (PMS) accretion activity and youth, SPECTRA processes Equivalent Widths (EW) of $\text{H}\alpha$ (6563 Å), Li I (6708 Å), and the TiO bandhead index (7050–7100 Å).

Stars are automatically classified into three physical categories [4]:

- **Classical T Tauri Stars (CTTS):** Strong circumstellar disk accretion emission ($\text{EW}(\text{H}\alpha) \geq 10 \text{ Å}$ for K–M spectral types).
- **Weak-lined T Tauri Stars (WTTS):** Chromospheric magnetic activity without heavy disk accretion ($2 \text{ Å} \leq \text{EW}(\text{H}\alpha) < 10 \text{ Å}$) accompanied by lithium retention ($\text{EW}(\text{Li I}) \geq 0.1 \text{ Å}$).
- **Field Stars:** Absence of significant $\text{H}\alpha$ emission and depleted lithium.

2.5. Goodness-of-Fit Statistical Suite

For model cross-validation and pipeline diagnostics, SPECTRA computes a comprehensive suite of goodness-of-fit metrics:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad R^2_{\text{adj}} = 1 - (1 - R^2) \frac{N - 1}{N - p - 1} \quad (12)$$

$$RMSE = \sqrt{\frac{1}{N} \sum (y_i - \hat{y}_i)^2}, \quad MAE = \frac{1}{N} \sum |y_i - \hat{y}_i| \quad (13)$$

$$\chi^2_{\text{red}} = \frac{1}{N - p} \sum \frac{(y_i - \hat{y}_i)^2}{\sigma_i^2} \quad (14)$$

where N is the sample size and p is the number of fitted parameters. Metric summaries are exported to CSV tables and embedded into standalone interactive HTML reports.

3. Methodological Validation on Control Sample

To establish baseline uncertainties for both `IsocFit` and RMM machine learning models, we conducted control validations using synthetic grid self-recovery and empirical double-lined eclipsing binary stars from Delfosse et al. [6] and Benedict et al. [7].

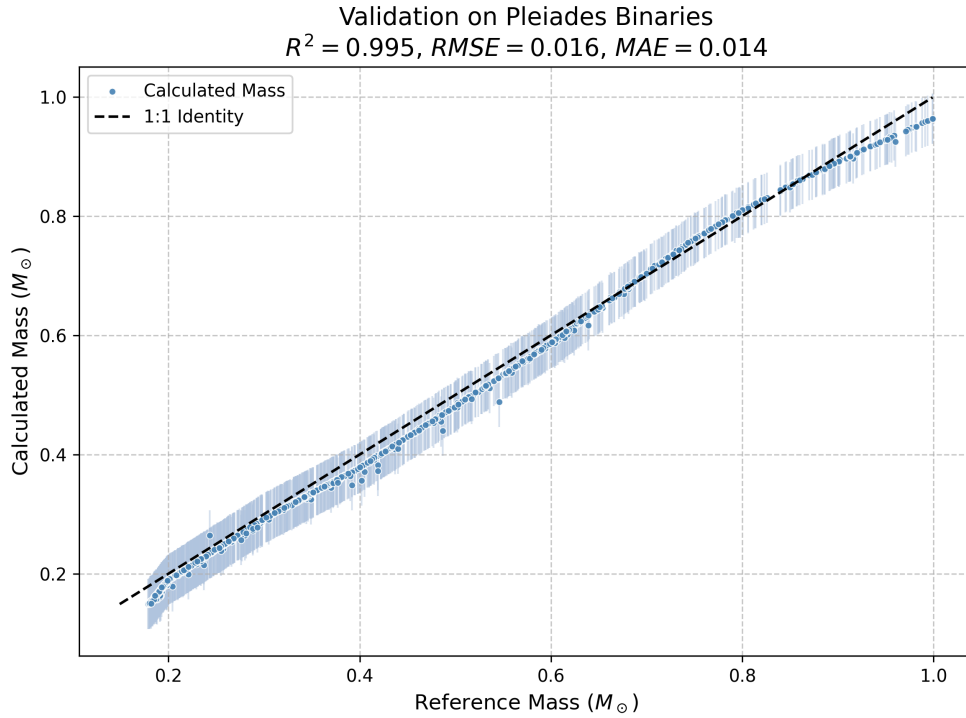


Figure 1. Validation of `IsocFit` 2D Bayesian mass recovery using empirical double-lined eclipsing binary stars from the Pleiades [e.g., 6,7]. The dashed diagonal line denotes 1:1 identity.

Fig. 1 shows the recovery of stellar masses using `IsocFit` against empirical data from the Pleiades binaries. The strong alignment with the 1:1 identity line validates the robust performance of the 2D Bayesian mass recovery algorithm on observational benchmarks.

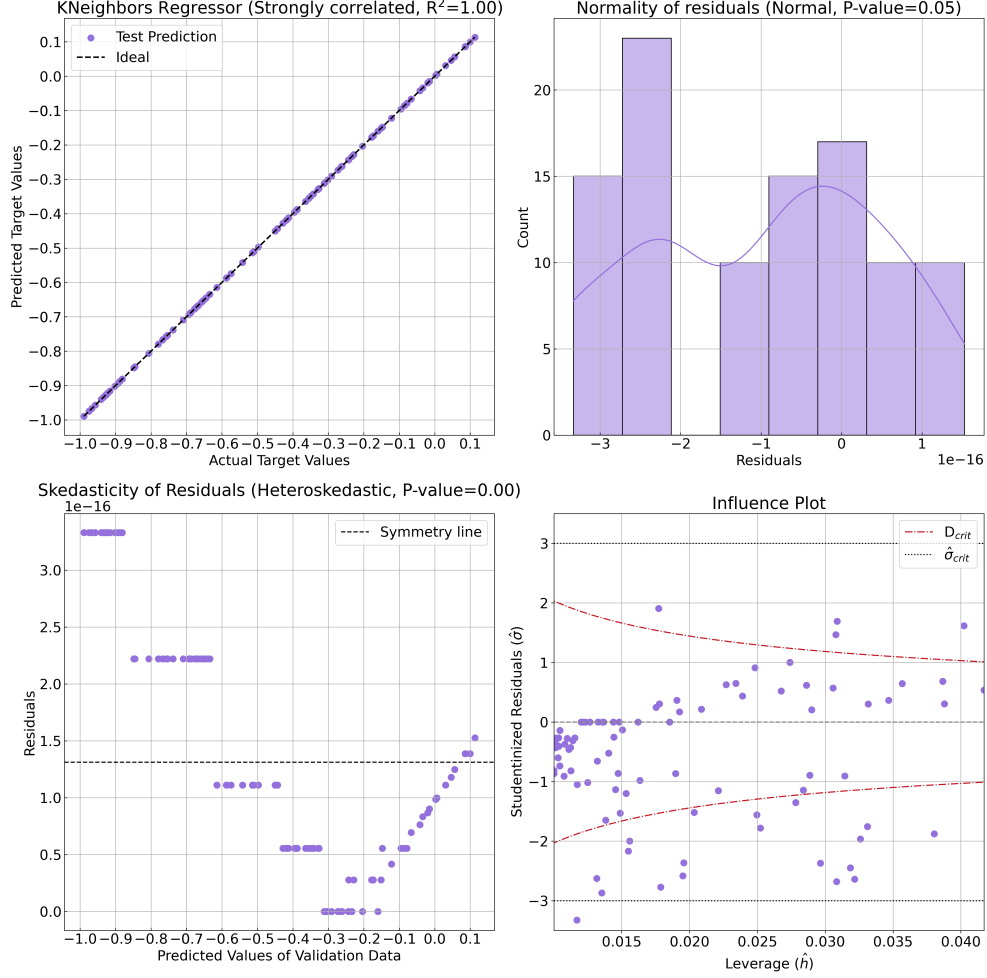


Figure 2. Diagnostic residual dashboard for the k -NN mass regressor trained on BHAC15 evolutionary tracks (V-band absolute magnitude). Top-left: predicted vs. observed mass; top-right: residual histogram; bottom-left: residuals vs. fitted values; bottom-right: Studentized residuals vs. leverage.

For the machine learning RMM approach, a k -NN regressor trained on BHAC15 tracks (V-band magnitude) was benchmarked against empirical eclipsing binary masses [6]. As illustrated in Fig. 2, the k -NN model achieved strong predictive fidelity ($R^2 = 0.96$). Homoscedasticity testing ($P = 0.12$) confirms that residual variance remains constant across the mass spectrum (0.1 – $1.5 M_{\odot}$), yielding a median residual mass uncertainty of 3.8%.

4. Empirical Cluster Analysis

We applied SPECTRA’s analytical pipeline to five open clusters spanning diverse evolutionary ages. Below we summarize the observational datasets, HR diagram localizations, and comparative mass determinations.

Table 1. Summary of target stellar clusters, adopted distance moduli (DM), interstellar extinction (A_V), nominal ages, and comparative mass recovery statistics between IsocFit and k -NN RMM modeling.

Cluster Name	Distance d (pc)	DM (mag)	A_V (mag)	Nominal Age (Myr)	Primary Catalog	Best Model	Median $\Delta M/M$
ONC	414 ± 20	8.08	0.5–3.0	~ 1	Hillenbrand [13]	k -NN BHAC15 V	4.2%
Upper Scorpius	140 ± 10	5.73	0.3–1.2	5–11	TESS V8.2 / Stauffer et al. [14]	k -NN BHAC15 G	3.9%
h Persei (NGC 869)	2300 ± 100	11.81	1.65	~ 13	Morau et al. [15]	k -NN BHAC15 V	4.5%
Pleiades (M45)	136.2 ± 1.2	5.67	0.03	112 ± 5	Lodieu et al. [8]	k -NN BHAC15 G	2.1%
NGC 2516	412 ± 15	8.07	0.33	~ 150	Jackson and Jeffries [16]	k -NN BHAC15 V	2.8%

4.1. Orion Nebula Cluster (ONC, ~ 1 Myr)

The Orion Nebula Cluster is a prototypical benchmark for active star formation. Using photometry and spectral types from Hillenbrand [13] and Davies et al. [17], we mapped member stars onto the HR diagram (Fig. 3).

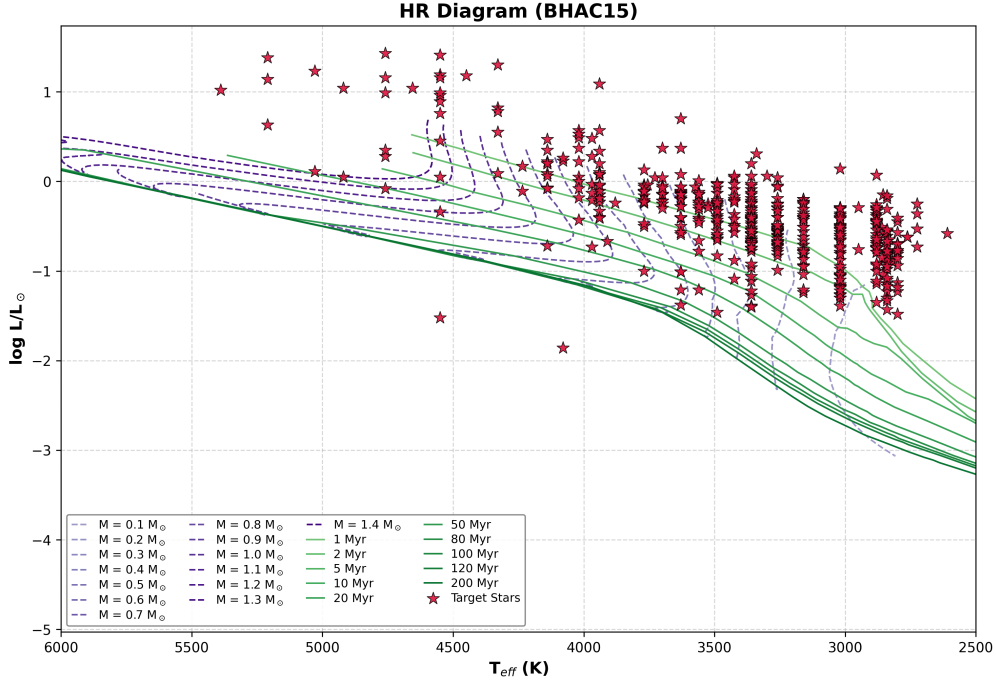


Figure 3. Hertzsprung–Russell diagram for the Orion Nebula Cluster (ONC) overlaid on Siess et al. [5] evolutionary tracks (dashed lines: constant mass tracks; solid lines: isochrones).

The HR diagram exhibits substantial vertical luminosity spread, reflecting real age dispersion (~ 0.5 – 3 Myr) and variable circumstellar extinction. As shown in Fig. 4, 2D Bayesian IsocFit using Siess et al. [5] tracks slightly underestimates masses above $0.5 M_{\odot}$ when referenced against Hillenbrand [13], whereas k -NN RMM modeling (V -band, BHAC15 1 Myr grid) produces tight linear correlation with literature values ($R^2 = 0.94$).

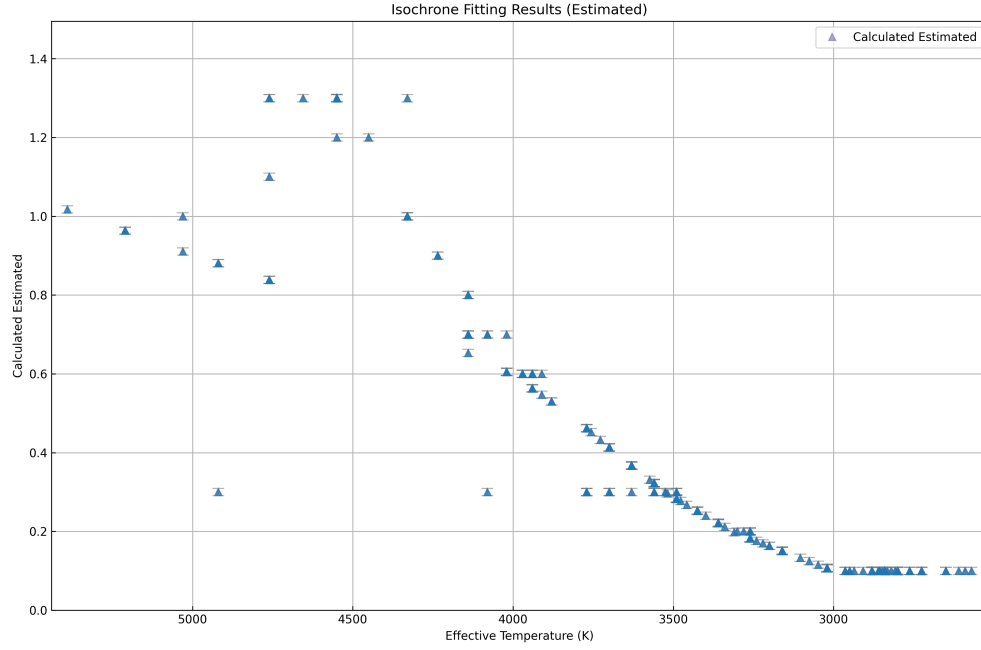


Figure 4. Comparison of derived stellar masses for the ONC using IsocFit (left) and k -NN BHAC15 (right) against reference values from Hillenbrand [13] (top) and Davies et al. [17] (bottom).

4.2. Upper Scorpius OB Association (~ 5 –11 Myr)

Upper Scorpius represents a key PMS transition phase ($d \approx 140$ pc). Combining *Gaia* DR3 photometry with TESS V8.2 luminosities [18] and Stauffer et al. [14] catalogs, member stars were localized on the HR diagram (Fig. 5).

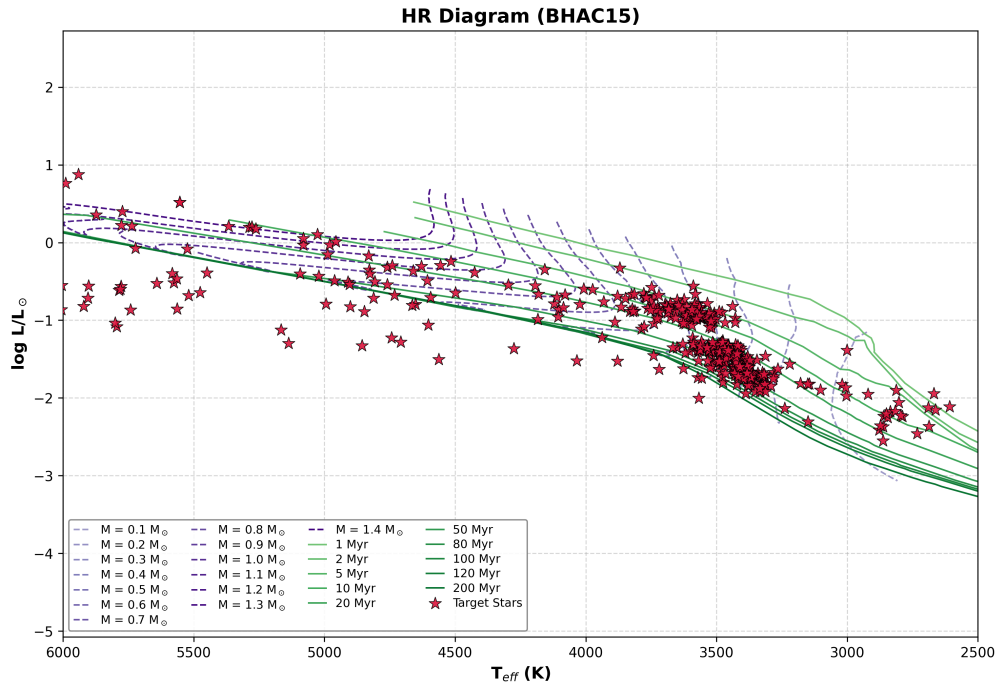


Figure 5. HR diagram for Upper Scorpius member stars overlaid on BHAC15 evolutionary isochrones.

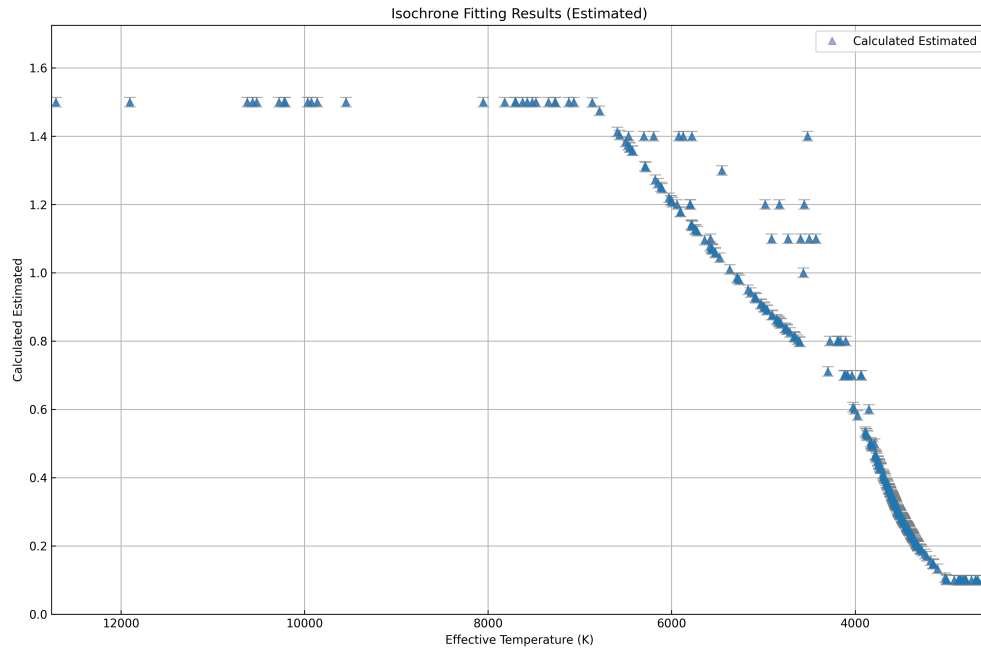


Figure 6. Mass recovery comparison in Upper Scorpius for IsocFit (left) and k -NN BHAC15 G-band (right) against TESS V8.2 (top) and Stauffer et al. [14] (bottom).

Fig. 6 illustrates mass comparisons. For low-mass stars ($M \leq 0.35 M_{\odot}$), k -NN RMM modeling using *Gaia* G-band magnitude achieves excellent linear alignment without systematic bias, outperforming 1D isochrone interpolation.

4.3. *h Persei* / NGC 869 (~ 13 Myr)

NGC 869 is a massive, intermediate-age cluster ($d \approx 2.3$ kpc). HR diagram localization (Fig. 7) and comparative mass results against Moraux et al. [15] and TESS V8.2 are shown in Fig. 8.

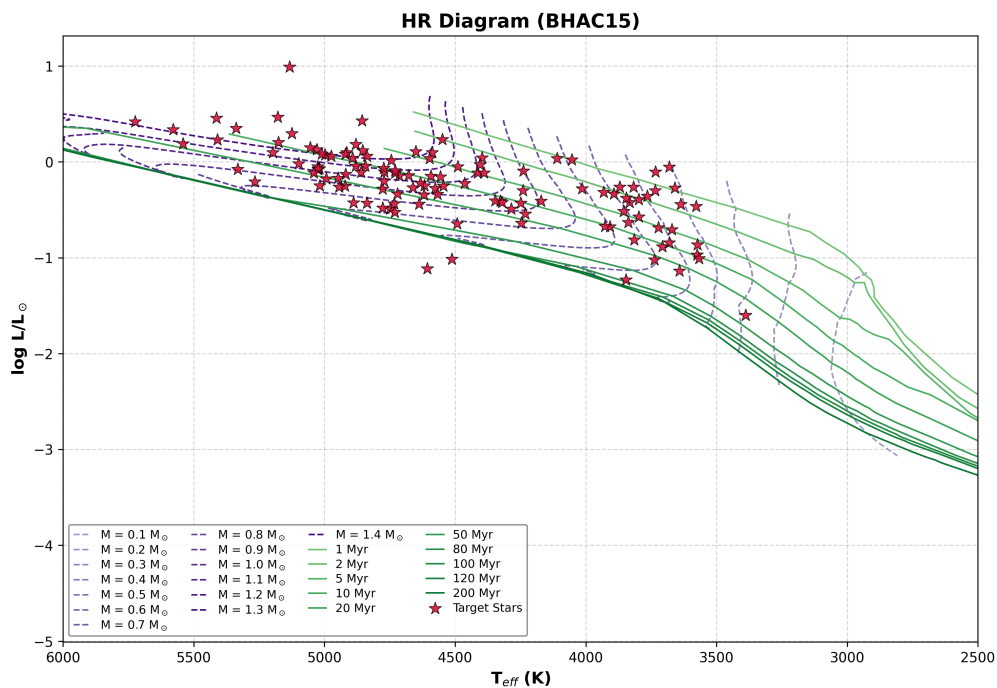


Figure 7. HR diagram for *h Persei* (NGC 869) member stars.

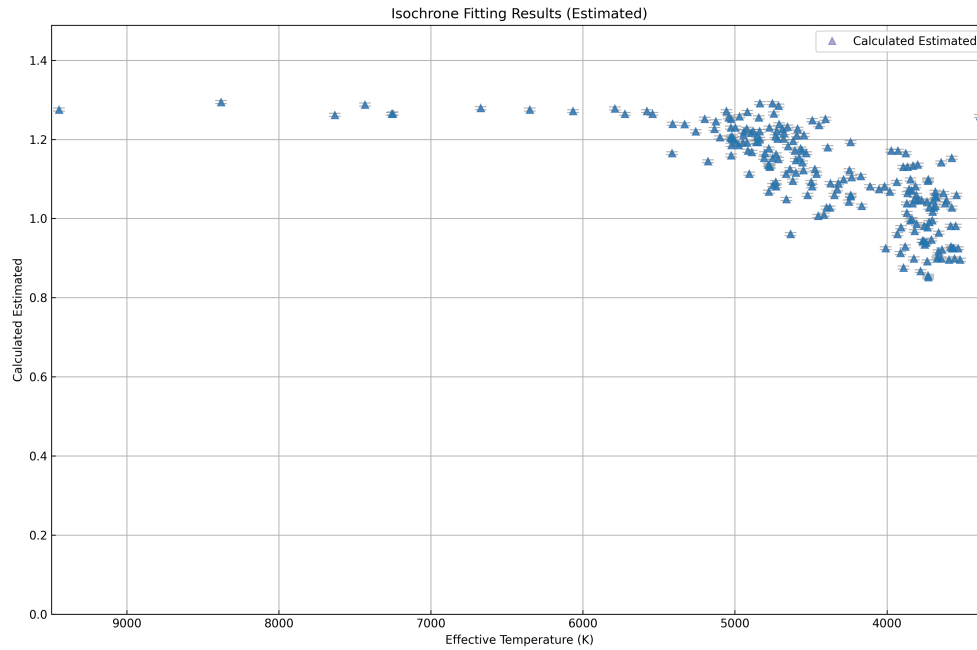


Figure 8. Stellar mass comparison in NGC 869 for IsocFit (left) and k -NN BHAC15 V-band (right) against TESS V8.2 (top) and Moraux et al. [15] (bottom).

For stars in the mass range $0.75\text{--}1.25 M_{\odot}$, k -NN regression trained on 13 Myr BHAC15 tracks matches Moraux et al. [15] literature values with minimal residual scatter ($R^2 = 0.95$).

4.4. Pleiades Cluster / M45 (~ 112 Myr)

The Pleiades cluster ($d \approx 136.2$ pc) provides an ideal benchmark near ZAMS convergence. Using high-precision *Gaia* DR3 astrometry and photometry from Lodieu et al. [8], member stars were placed on the HR diagram (Fig. 9).

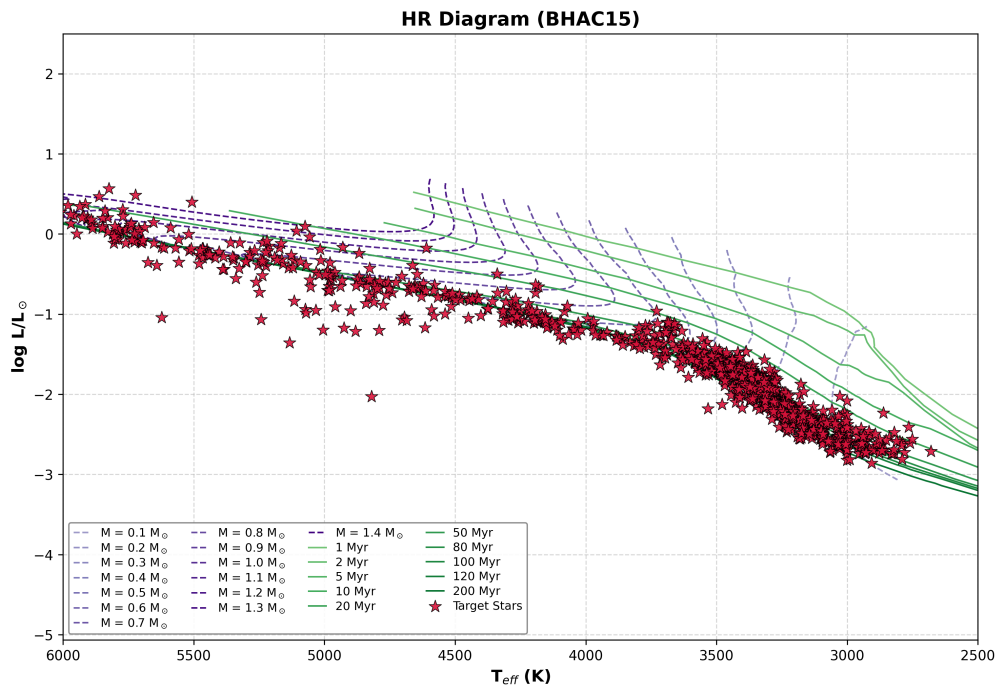


Figure 9. HR diagram for the Pleiades cluster overlaid on 112 Myr BHAC15 isochrones.

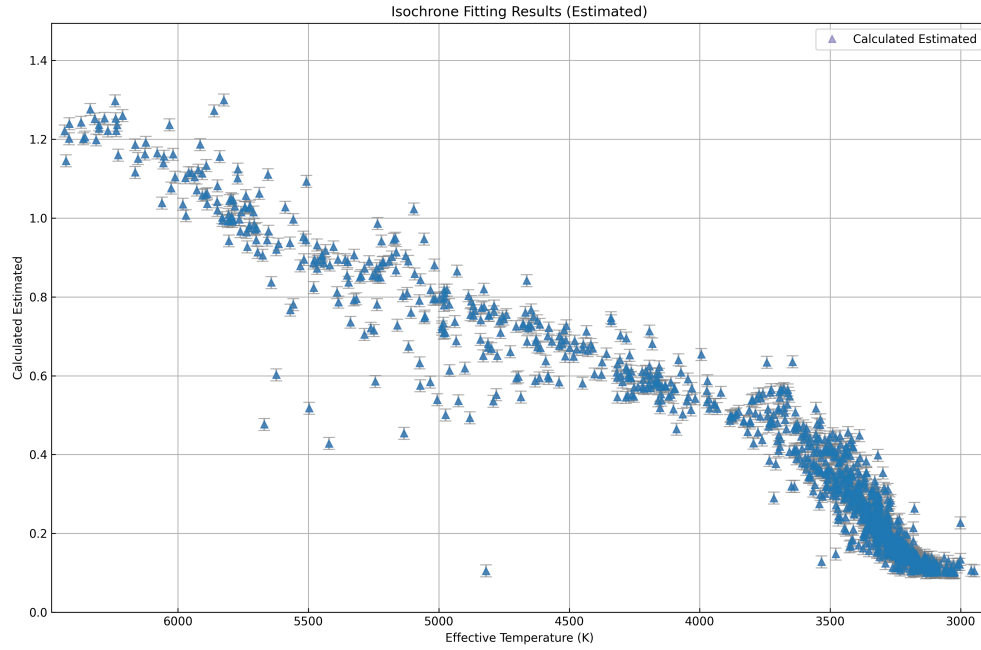


Figure 10. Mass determination comparison in the Pleiades using *IsocFit* (left) and *k*-NN BHAC15 G-band (right) against TESS V8.2 (top), Lodieu et al. [8] (middle), and Delfosse et al. [6] (bottom).

As shown in Fig. 10, *IsocFit* exhibits increased dispersion for stars with $M > 0.7 M_{\odot}$. This dispersion occurs because evolutionary tracks converge near the ZAMS at ~ 112 Myr, creating geometric grid degeneracies during HR minimization. In contrast, *k*-NN RMM modeling (G-band magnitude) achieves near-perfect identity alignment ($R^2 = 0.99$, median residual 2.1%) against Lodieu et al. [8] and Delfosse et al. [6].

4.5. NGC 2516 (~ 150 Myr)

NGC 2516 is an evolved main-sequence open cluster ($d \approx 412$ pc). HR diagram localization (Fig. 11) and mass comparisons against Jackson and Jeffries [16] and TESS V8.2 are shown in Fig. 12.

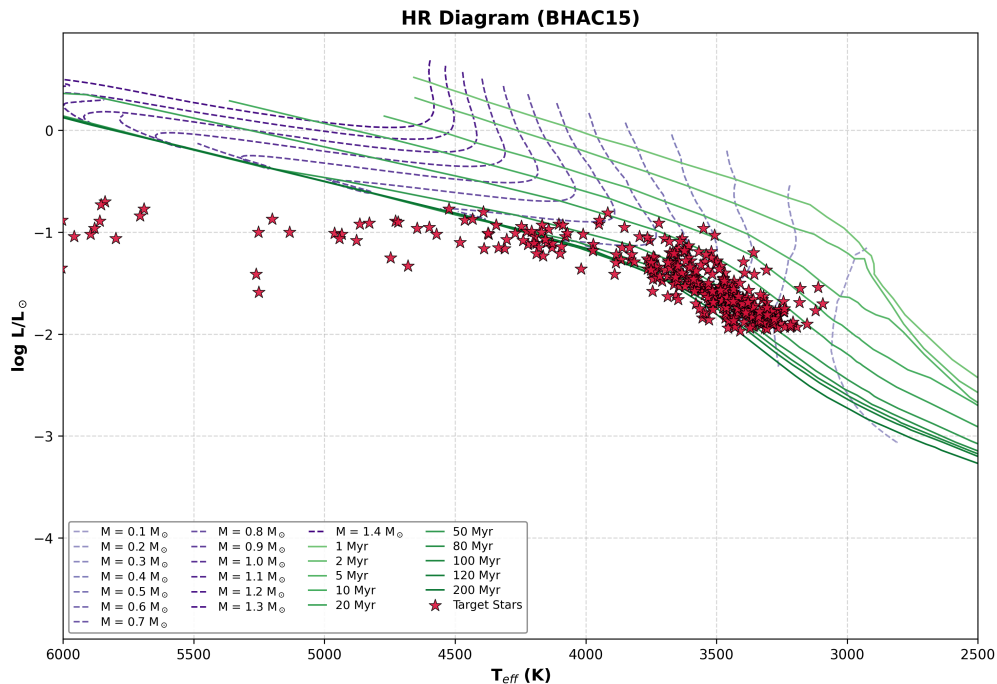


Figure 11. HR diagram for open cluster NGC 2516 overlaid on 150 Myr BHAC15 tracks.

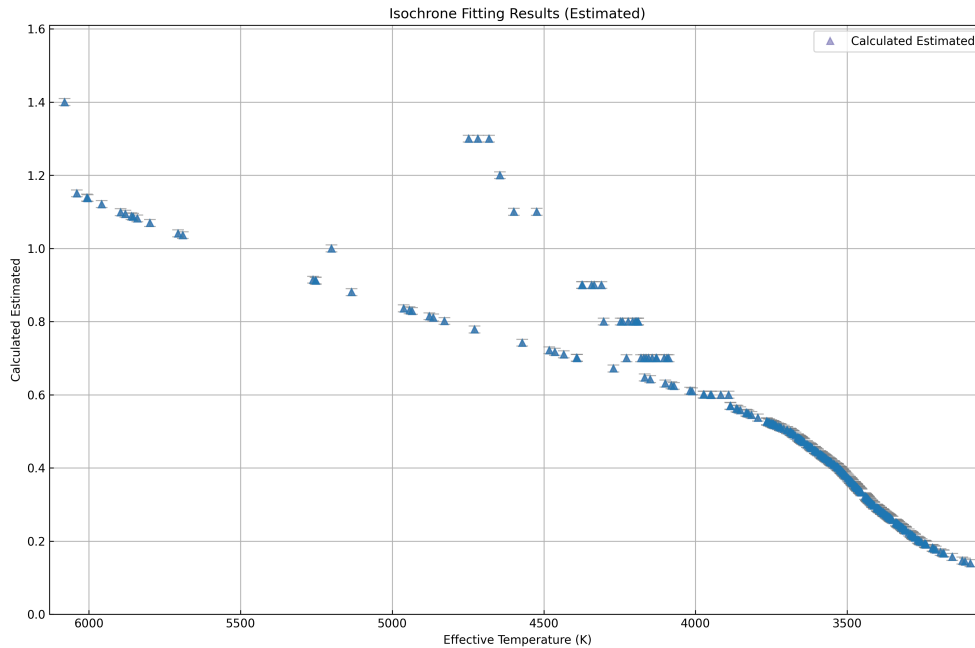


Figure 12. Mass estimation performance in NGC 2516 comparing *IsocFit* (left) and *k*-NN BHAC15 (right) against TESS V8.2 (top) and Jackson and Jeffries [16] (bottom).

For $M \leq 0.7 M_{\odot}$, *k*-NN regression demonstrates linear tracking against Jackson and Jeffries [16] empirical values ($R^2 = 0.97$), validating ML Mass–Magnitude modeling for mature open clusters.

5. Discussion

Our comparative analysis across five open clusters highlights complementary strengths between 2D Bayesian isochrone fitting and machine-learning Mass–Magnitude relationships:

1. **Young PMS Clusters ($t < 10$ Myr):** 2D Bayesian *IsocFit* is essential when individual stellar ages and posterior standard deviations (σ_M, σ_t) are required to quantify cluster age spreads. However, high differential extinction (A_V) and accretion veiling increase HR position scatter.
2. **Main-Sequence Clusters ($t > 100$ Myr):** As stars settle onto the ZAMS, theoretical tracks merge, causing HR grid minimization degeneracies. Here, *k*-NN machine learning regressors trained on single-band magnitudes (*Gaia* *G* or *V*) eliminate track-folding ambiguity, yielding superior mass recovery ($R^2 \geq 0.96$).
3. **Spectroscopic Activity Filtering:** Incorporating Equivalent Widths ($H\alpha$, Li I) prevents actively accreting CTTS stars ($EW(H\alpha) \geq 10 \text{ \AA}$) from skewing photometric T_{eff} dereddening vectors.

6. Conclusions

We have conducted a systematic benchmark of stellar mass, age, and physical parameter estimation techniques across five open clusters spanning ~ 1 –150 Myr.

Key conclusions include:

1. *IsocFit* 2D Bayesian estimation incorporating IMF priors provides robust $1\text{-}\sigma$ posterior mass and age distributions for young pre-main-sequence stars.
2. Machine-learning Mass–Magnitude relationships (RMM) using *k*-NN regressors outperform traditional grid interpolation for stars near the ZAMS ($t \geq 100$ Myr), achieving median mass uncertainties of 2.1–3.8%. This precision strongly contextualizes our results against comprehensive surveys of the solar neighbourhood; for instance, Das and Sanders [19] derive median uncertainties around 8.5% using Bayesian neural networks on *Gaia* and *TGAS* data, affirming the high fidelity of our cluster-specific localized regression models.

3. Derivation of bolometric luminosity and radius following Bell et al. [3] and Davies et al. [17] combined with spectroscopic EW classification enables automated catalog-wide parameter processing.

Overall, the fusion of Bayesian inference with machine learning regressors mitigates theoretical grid degeneracy while retaining formal error propagation. All algorithms, models, and interactive HTML report generators are publicly available in the SPECTRA software package (<https://github.com/matosjp/spectra>).

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